Comparative study of nonparametric density estimators.

Abstract

Nonparametric density estimations have been widely applied in economics and finance, but few comparative studies exist. The objective of this paper is to assess the performance of major density estimators by fitting univariate densities with heavy tails, which are relevant to financial application. We consider four approaches: a fixed bandwidth kernel estimator of Sheather and Jones (1991), the adaptive kernel estimator of Silverman (1986), the Hermite series (SNP) method of Gallant and Nychka (1987), and the logspline approach of Kooperberg and Stone (1991, 1992). Our Monte Carlo study indicates that the adaptive kernel and logspline methods are superior to the others. The performance of the SNP and logspline methods are compared further through evaluation of the numerical rate of decreasing Hellinger error as the number of parameters increases in approximating the true Student-t densities. The derived rate for the SNP method is shown to he consistent with existing literatures.

1. Introduction

Nonparametric density estimations have been widely applied in economics and finance. Without rigid assumptions about the form of the distribution, the methods flexibly exploit the information from the data for capturing the stylized facts or for judging the model fitting. When densities are heavy-tailed, however, the shape of the estimates can differ depending on the chosen methods. In application to finance, it is important to use a method robust for heavy-tailed densities, since the distributions of asset price change tend to have heavy tails as was first reported by Fama (1965), and information about sudden and large price change is contained in the tails.

The primary objective of this paper is to assess the performance of major density estimators by fitting univariate densities with heavy tails. We consider four approaches: a fixed bandwidth kernel (fixed kernel) estimator of Sheather and Jones (1991), the adaptive bandwidth kernel estimator of Silverman (1986), the Hermite series (SNP) method of Gallant and Nychka (1987), and the logspline approach of Kooperberg and Stone (1991, 1992).

The Monte Carlo simulation study in this paper indicates that the logspline and adaptive kernel methods are superior to the other methods in estimating densities with heavy tails. We analyze this result further by numerically evaluating the performance of the SNP and logspline estimators in approximating the true densities, and derive the rates of error reduction for Student-t densities with degrees of freedom, ν. As a function of the number of parameters, p, the Hellinger error in the logspline estimation is found to decrease faster than p^{-7}, possibly with exponential law, $e^{-\tilde{\alpha}p^{\tilde{\beta}}}$, that is independent of tail heaviness. On the other hand, the Hellinger error of the SNP method decreases algebraically, $p^{-\nu/2}$ where ν means tail heaviness. The derived rate for the SNP method is shown to he consistent with existing literatures.

2. The Methods

This section summarizes the algorithms of density estimators studied in this paper. For samples $X_1, X_2, \ldots, X_n$, kernel based density estimator has the form

$$\hat{f}_{FK}(x) = \frac{1}{n}\sum_{i=1}^{n} \phi\left(\frac{x - X_i}{h}\right), \tag{1}$$

where $\phi(\cdot)$ denotes the standard Gaussian kernel, and a fixed bandwidth h controls the degree of smoothing. In selecting h, we consider two-stage direct plug-in bandwidth selection method of Sheather and Jones (1991), which is shown to perform quite well for many density types by Park and Turlach (1992) and Wand and Jones (1995).

The adaptive kernel density estimator of Silverman (1986) is defined as

$$\hat{f}_{\mathrm{AK}}(x) = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{h\lambda_i}\phi\left(\frac{x - X_i}{h\lambda_i}\right). \tag{2}$$

Here, λ_i is the local bandwidth factor which makes the bandwidth $\lambda_i h$ to be narrower around modes and wider around tails. Based on the pilot estimate by fixed kernel method, $\tilde{f}(x)$, λ_i is determined as $\lambda_i = \tilde{f}(X_i)/g^{-\alpha}$, where $\log g = n^{-1}\sum \log \tilde{f}(X_i)$ with $\alpha = 1/2$. The bandwidth of the pilot estimate is determined by Silverman (1986)'s rule-of-thumb, $h = 0.9An^{-1/5}$ where $A = \min\{\text{standard deviation, interquartile range}\}/1.34$.

The Gallant and Nychka (1987) SNP approach is a truncated Hermite series estimator defined as

$$\hat{f}_{\text{SNP}}(x,\theta) = \left[\sum_{i=0}^{K} \theta_i \bar{H}_{e_i}(x)\right]^2 e^{-x^2/2} + \varepsilon_0 \phi(x), \quad \theta = (\theta_0, \ldots, \theta_K), \quad \sum_{i=0}^{K} \theta_i^2 + \varepsilon_0 = 1, \quad (3)$$

where $\bar{H}_{e_i}(x)$ is the normalized Hermite polynomial term, $\phi(x)$ is the standard normal density, and ε_0 is small positive number. Estimation is by quasi-maximum likelihood:

$$\hat{\theta} = \arg\max_{\theta,\mu,\sigma} \frac{1}{\text{n}} \sum_{\text{j}=1}^{\text{n}} \log\left\{\frac{1}{\sigma}\, \hat{f}_{\text{SNP}}\left[\frac{X_j - \mu}{\sigma}\right]\right\}, \quad (4)$$

where μ and σ^2 correspond to mean and variance. The truncation point is either at $K = n^{1/5}$ or according to AIC or BIC model selection criterion. The number of parameters is $p = K + 3$, due to inclusion of μ, σ, and the 0-th order term.

The logspline method of Kooperberg and Stone (1991, 1992) models a log-density function as a cubic B-spline:

$$\hat{f}_{\text{LS}}(x,\theta) = \exp\left\{\sum_{i=1}^{p} \theta_i B_i(x) - c(\theta)\right\}, \; x \in \mathcal{R}, \quad (5)$$

where $c(\theta)$ is a normalization factor. Then the coefficients θ are estimated by maximizing log-likelihood function given by

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \log\left(\hat{f}_{\text{LS}}(X_i,\theta)\right). \quad (6)$$

Final model selection is based on AIC or BIC.

3. Monte Carlo Simulation Study

This section compares the performance of the four density estimators in fitting heavy-tailed densities via Monte Carlo study. In addition to the normal density, $t(3)$ and $t(1)$ densities are chosen to see the effect of tail behavior. The lognormal density is included as different type of heavy tail. All the densities are rescaled so that 99.9% quantiles are included in the range of $[-3, 3]$. The performance is evaluated by the Hellinger error between the true density f and the estimated density $\hat{f}$ defined as

$$\text{Hellinger}(\hat{f}, M) = \text{E}\int_{-M}^{M} \left\{\sqrt{\hat{f}(x)} - \sqrt{f(x)}\right\}^2 dx. \quad (7)$$

The Hellinger error emphasizes the error around the tail part. We set $M = 3$.

The source codes used for the comparative study are as follows: Codes for the SNP estimation are by Fenton and Gallant (1996a), and those for the logspline method are by Kooperberg and Stone (1991, 1992). The adaptive kernel estimator is programmed by Roger Koenker based on Silverman (1986). The direct plug-in fixed kernel method is coded by Wand and Jones (1995). In the Monte Carlo simulation study, the default settings are used for all the algorithms. The logspline program includes knot deletion procedure. The numbers of parameters p for the SNP and logspline estimators are determined so that BIC is minimized.[1]

Table I summarizes error statistics for all the densities studied. The sample size is fixed at 500. The error statistics are the average of 500 replications. The errors of the logspline and adaptive kernel methods are relatively similar in size for all the densities with different tail heaviness, while those of the SNP and fixed kernel methods increase as the tails become heavier. The relative performance for heavy tails measured by the Hellinger error is better in the order of 1) logspline, 2) adaptive kernel, 3) fixed kernel, and 4) SNP. In particular, the performance of the SNP estimator relative to that of the logspline estimator becomes significantly worse as the tail of the density becomes heavier.

To assess the qualitative performance in fitting heavy-tailed densities, $t(3)$ density is estimated using the four methods for one realization of random sample, and the results are illustrated in Figure 1. The dotted lines indicate the true $t(3)$ density while the solid lines denote the estimates. To emphasize the tail fitting performance, the vertical axis is transformed to be log-scaled in Figure 1. The SNP estimate around tails involves high-frequency oscillations. This is probably because the SNP estimator has no other way to fit heavy tails than mixing higher degree of Hermite polynomials with Gaussian tail.[2] [3] The fixed kernel estimate is also problematic especially in tails. The possible reasons are 1) it cannot produce estimates where there is no data, 2) the fixed bandwidth becomes very narrow to fit sparse data around tails, making generated estimates include many small bumps because just one kernel is put for one sample. The estimates of the adaptive kernel and logspline methods do not have significant problem in tail fitting of $t(3)$ densities. The particularly good performance of the logspline estimator in the log scaled graph might partly because it optimizes with respect to the log-density, and extrapolates linear intervals in log scale beyond the extreme knots.

The density estimates of the daily change of S&P500 index are compared in Figure 2, where only the vertical axis is in log scale. The data period is from 1989 to 1998,

[1] The upper bound of the Hermite polynomial order K is set as $K = n^{1/2}$ to avoid non-convergence.

[2] The leading term can be replaced from Gaussian to ARCH or GARCH with Gaussian tails, which might improve the tail fitting performance.

[3] The technique to remove the small bumps in the tail as in Efromovich (1999), section 3.1 might also improve the performance.

containing 2528 data. The daily price change, Δp is defined as $\Delta p = \log p_t - \log p_{t-1}$, where p_t denotes the price at time t. In Figure 2, as the case of $t(3)$ density estimation, we observe the same pattern of differences in the tail shape of the estimates depending on the chosen method, suggesting the lobes in the tails of the fixed kernel and SNP method are likely to be artifacts.

There is a dispute about the existence of the side lobes in the tails of the densities of financial asset price change. Based on the SNP density estimation of daily pound/dollar data from 1974 to 1983, Gallant, Hsieh, and Tauchen (1991) found the side lobes in the error density of ARCH specification, and Engle and Gonzalez-Rivera (1991) report similar lobes in the distribution of the standardized residuals from a GARCH(1,1) model. They claim that the detected side lobes are the remained nonlinearity which cannot be explained by the ARCH/GARCH formulation. However, the study by Bolleslev (1987) and Balillie and Bollerslev (1989) on analyzing data for the British pound for 1980-1985, found little evidence against GARCH(1,1) with t-distributed errors. Bollerslev, Chou, and Kroner (1992) suggest that this particular pattern is influenced by a few outliers, therefore a feature for the specific period. Considering the artificial bumps in the tails of estimate of $t(3)$ density, we see the detected hump-shaped tail in the pound/dollar data is most likely be an artifact of the SNP method.

4. The Approximation Rate of the SNP Method in Estimating Heavy-Tailed Densities

This section numerically evaluates the performance of the SNP estimator in approximating the true densities, and derives the rate of convergence of the SNP estimator in estimating $t(\nu)$ densities, where ν is the degree of freedom. The SNP method models a density as a truncated Hermite series, and the approximation error decreases as the number of the Hermite polynomial terms, K, increases. The coefficient of the last Hermite polynomial term, θ_K, also decreases as K increases, conveying information about the error at K. In approximating the true densities, we determine θ by orthogonalization rather than by maximum likelihood as the original SNP formulation of Gallant and Nychka (1987). First we show how θ is connected to the Hellinger error. Next, based on the rate θ_K decreases as K grows, we investigate the rate that Hellinger error decreases as the number of parameters increases. Finally, the derived result is compared with existing SNP rate results.

4.1 The coefficients of Hermite polynomial terms and the Hellinger error

Under certain regularity conditions, the true density f can be represented as an infinite sum of normalized Hermite polynomials

$$f(x,\theta) = \left[\sum_{i=0}^{\infty} \theta_i w_i(x)\right]^2, \quad \theta = (\theta_0, \theta_1, \ldots\ldots), \quad \sum_{i=0}^{\infty} \theta_i^2 = 1. \tag{8}$$

The coefficients θ_i can be determined by calculating

$$\theta_i = \int_{-\infty}^{\infty} \sqrt{f(x,\theta)}\ w_i(x)dx. \tag{9}$$

We will now show that θ_i in (9) minimize Hellinger error. Let $T_K(x,\lambda)$ be a partial sum of normalized Hermite polynomials

$$\sqrt{T_K(x,\lambda)} = \sum_{i=0}^{K} \lambda_i w_i(x), \tag{10}$$

where λ_i are arbitrary. Then the Hellinger error between $f(x)$ and $T_K(x,\lambda)$,[4] $H(f,T_K)$, is transformed as

$$H(f,T_K) = \int_{-\infty}^{\infty} \left\{ \sqrt{f(x)}\ - \sqrt{T_K(x,\lambda)} \right\}^2 dx = 1 + \sum_{i=0}^{K} (\lambda_i - \theta_i)^2 - \sum_{i=0}^{K} \theta_i^2. \tag{11}$$

So $H(f,T_K)$ is minimized over the λ_i's by choosing $\lambda_i = \theta_i : i = 1,\ldots,K$.

Having chosen the λ_i's this way we have

$$H(f,T_K) = \sum_{i=K+1}^{\infty} \theta_i^2, \tag{12}$$

so ε_0 in (3) thus equals to the Hellinger error.

4.2 Evaluating approximation error of the SNP estimator.

Let p be the number of parameters, where $p = K+3$. We computed θ_p up to $p = 200$ using (9) for densities with varying tail behavior: Student-t distributions with degree of freedom $\nu = 1$, 2, 3, 5, 11, and bimodal density.[5] All the densities but $t(1)$ and $t(2)$ are standardized so that their variance equal to unity.[6] In Figure 3, for each of the densities, the logarithm of absolute θ_p is plotted as a function of p in log scale. The plotted curves are asymptotically straight for the Student-t densities, and curved downward for the bimodal density. Moreover, as the degrees of freedom ν increases, the plotted slope becomes flatter, indicating that the rate of approximation error reduction measured by the Hellinger error becomes slower.[7]

We hypothesize that for algebraic tailed densities

$$\theta_p = Ap^{\alpha}. \tag{13}$$

[4]Exactly speaking, $f_K(x,\lambda) = T_K(x,\lambda) + \varepsilon_0\phi(x)$ should be used instead of $T_K(x,\lambda)$. However, the term $\varepsilon_0\phi(x)$ in (3) contributes for normalization, but is not effective for reducing error. Thus neglecting the term $\varepsilon_0\phi(x)$ for sufficiently large K may be justified.

[5]A mixture of normal such that $\frac{1}{2}N[-1,0.5^2] + \frac{1}{2}N[1,0.5^2]$.

[6]This is because the SNP model in this section does not automatically standardize the estimated densities, while the original SNP model automatically does.

[7]The Hellinger error in this section is evaluated in the range of x from $-\infty$ to ∞, in contrast to the previous section.

Suppose (13) holds and consider approximating the sum $H(f, T_K) = \sum_{i=K+1}^{\infty} \theta_i^2$ by the integral $h(p) = \int_p^{\infty} \theta^2(v)dv$. So, for (13) we have,

$$h(p) = \int_p^{\infty} \theta^2(v)dv = A^2 \int_p^{\infty} v^{-2\alpha} dv = A^2(2\alpha - 1)v^{-(2\alpha-1)}. \tag{14}$$

The value for α corresponds to the slope of the lines for Student-t densities in Figure 3. The least squared regression indicates the relation

$$\alpha = -(\nu + 2)/4, \tag{15}$$

for every ν studied. Based on the conjecture (15), we obtain the rate that the Hellinger error decreases as parameter grows,

$$h(p) \sim p^{-\nu/2}. \tag{16}$$

If $p = n^{\alpha}$ is assumed, the Hellinger rate of convergence becomes

$$h(p) \sim \sum_{i=p}^{\infty} \theta_i^2 \sim p^{-\nu/2} \sim n^{-\nu\alpha/2}. \tag{17}$$

4.3 Discussion

There are two results for the rate of error reduction of the SNP estimator: One is by Fenton and Gallant (1996b) with respect to the L_1 distance, and the other is by Coppejans and Gallant (2000) with respect to the Hellinger distance. These results are restricted to certain densities satisfying some tail conditions. Note that the determination of the Hermite coefficient is by maximum likelihood. In particular, if $p = n^{\alpha}$ is assumed, the best Hellinger error rate in Coppejans and Gallant (2000) is achieved when $\alpha = \frac{1-\delta}{k+1}$, which gives, $h(p) \sim n^{-k(1-\delta)/2(k+1)} = n^{-k\alpha/2}$.

Our numerical Hellinger rate can be shown to be consistent with their rates by examining a Lemma used as a key component in their proof.

Lemma (Fenton and Gallant, 1996b) *If g is k-times differentiable with $g^{(j)} \in L_2(e^{-x^2/2})$ for $j = 0, 1, \ldots, k$, then the Hermite coefficients of g satisfy $\sum_{i=p}^{\infty} \theta_i^2 = o(p^{-k})$, where $g^{(j)} \in L_2(e^{-x^2/2})$ if $\left\{ \int_{-\infty}^{\infty} \left[g^{(j)}(x) \right]^2 e^{-x^2/2} \right\}^{1/2} < \infty$.*

Here, the k-times differentiability of g is not restrictive, but L_2 differentiability of g is a tight condition especially for heavy-tailed densities. In the case of standard $t(\nu)$-density whose variance is $\nu/(\nu-2)$ for $\nu > 2$, we analytically computed the maximum value of k for each of 11 degree of freedom ν from $\nu = 1$ to 11. The following relation is found to hold: $k = \nu/2$ for even ν, and $k = (\nu - 1)/2$ for odd ν, justifying to approximate the maximum order of differentiability, k, as

$$k \sim \nu/2. \tag{18}$$

Therefore, our result $\sum_{i=p}^{\infty} \theta_i^2 \sim p^{-\nu/2}$ in (17) becomes consistent with their Lemma $\sum_{i=p}^{\infty} \theta_i^2 \sim p^{-k}$.

By replacing k with $\nu/2$, we can rewrite the rate of convergence results of Gallant and Coppejans (2000) as a function of ν in estimating $t(\nu)$ densities:

$$h(p) \sim n^{-k\alpha/2} \sim n^{-\nu\alpha/4}, \tag{19}$$

which is slower than our Hellinger rate (17). The rate of convergence implied either by (17) or by (19) is shown to be considerably slow and dependent of the tail heaviness.

5. Comparison of the Approximation Rate of the SNP Method with the Logspline Method

We briefly summarize the approximation rate of the logspline method as a function of the number of parameters, which can be compared with the result of the SNP method described in the previous section.

The logspline approximation rate is derived by modifying the logspline codes (Kooperberg and Stone, 1991, 1992): the summation in (5) is replaced with integration so that the true density can be approximated, and the effect of knot placement on the rate of convergence is kept as minimum. The Hellinger error of the logspline method is found to decrease faster than p^{-7}, possibly with exponential law, $e^{-\tilde{\alpha}p^{\tilde{\beta}}}$, that is independent of tail heaviness. This is much faster rate than that of the SNP method, $p^{-\nu/2}$, which becomes slower as tail becomes heavier.

Figure 4 summarizes the result of this comparative study for the case of $t(3)$ density. The solid line is the Hellinger error for the SNP method as a function of the number of parameters p derived in section 4, and the dotted line is the counterpart for the logspline method. Both horizontal and vertical axes are in log scale. The white and black bullets are the Monte Carlo simulation results for the logspline method and the SNP method, respectively. The Monte Carlo results for the adaptive and fixed kernel methods are also added for reference in the right side of the figure, of which the numbers of parameter are not defined and illustrated arbitrary in the horizontal axis. It is observed that the Hellinger approximation error of the logspline method decreases significantly faster than that of the SNP method, and the rate is consistent with the Monte Carlo simulation results: average of 500 replications for the cases of $n = 100$, 500, and $1,000$.

6. Conclusion

It is clear that for heavy-tailed densities like Student-t, the rate of convergence of the SNP estimator is remarkably slow compared to logspline or even conventional kernel methods. Since financial data are often characterized as heavy-tailed, it seems prudent to consider the logspline method or adaptive kernel method for financial applications.

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Table I: Simulation Results for Densities with Heavy Tails
(Sample size 500, average of 500 simulated samples)

	Method	**Normal** mean	(s.d.)	**t(3)** mean	(s.d.)	**t(1)** mean	(s.d.)	**Lognormal** mean	(s.d.)
Hellinger	F.Kernel	0.0035	(0.0017)	0.0112	(0.0026)	0.1344	(0.0061)	0.0237	(0.0032)
	A.Kernel	0.0035	(0.0016)	0.0049	(0.0020)	0.0247	(0.0038)	0.0121	(0.0022)
	SNP	0.0010	(0.0010)	0.0111	(0.0069)	0.4242	(0.2946)	0.0383	(0.0168)
	Logspline	0.0029	(0.0016)	0.0045	(0.0026)	0.0087	(0.0038)	0.0052	(0.0025)
p	SNP	3.0	(0.0)	10.9	(4.2)	25.3	(1.5)	22.9	(2.9)
	Logspline	4.2	(0.6)	6.2	(0.5)	9.4	(0.8)	5.8	(1.0)

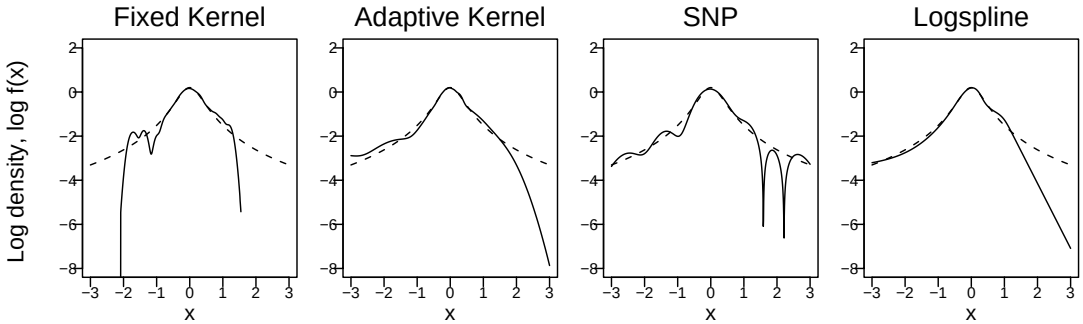

Figure 1: Estimates of $t(3)$ Densities Given a Sample with Size 500. Dotted lines indicate the true $t(3)$ density, while the solid lines denote the estimates.

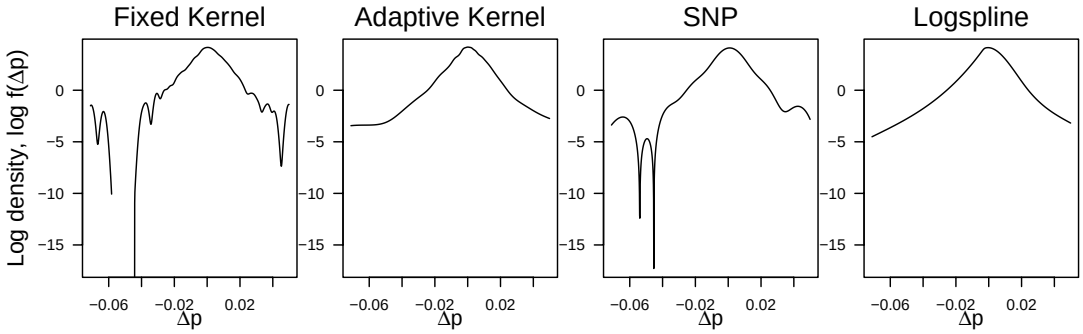

Figure 2: Density Estimates of Daily Change of S&P500 Index (1989-1998). Sample size is 2528.

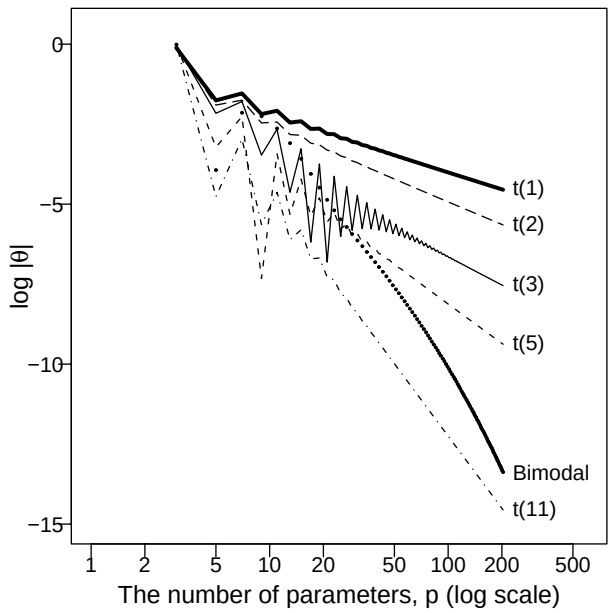

Figure 3: The Coefficients of Hermite Polynomial Terms of the SNP Estimates, θ, as a Function of the Number of Parameters in Approximating the True Densities.

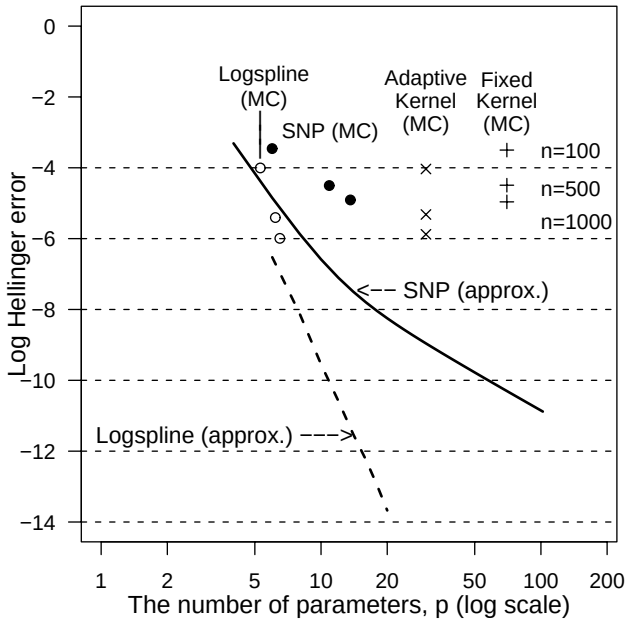

Figure 4: The Hellinger Error as a Function of the Number of Parameters in Approximating or Estimating $t(3)$ Density. Points labelled as "MC" indicate the results from the Monte Carlo study of estimating $t(3)$ density. Lines labelled as "approx." indicate the approximation error in fitting the true $t(3)$ density.